MAPS BETWEEN PROJECTIVE VARIETIES

In this lecture, we will cover some basic theory of algebraic maps between projective algebraic varieties. We will not prove all the things covered here as this and the next few lectures are to be taken more like a glossary of terms and results that we will use. The students are expected to work out the details on their own, as this will give them much needed experience with these objects.

1. Rational maps and morphisms

Let K be a perfect field and $\bar{K}$ be a fixed algebraic closure of K with $G_K = \mathrm{Gal}(\bar{K}/K)$.

Definition 1.1. Let $V_1 \subset \mathbb{P}^m(\bar{K})$ and $V_2 \subset \mathbb{P}^n(\bar{K})$ be projective varieties. A *rational map from V_1 to V_2* is an equivalence class of $(n+1)$-tuples

$$(f_0, \ldots, f_n), \qquad f_i \in \bar{K}(V_1),$$

which are not all zero and have the property that, for every point $P \in V_1$ at which $f_0, \ldots, f_n$ are all regular and not all of $f_0(P), \ldots, f_n(P)$ are zero, we have

$$[f_0(P), \ldots, f_n(P)] \in V_2.$$

Two such tuples $(f_0, \ldots, f_n)$ and $(g_0, \ldots, g_n)$ are equivalent if

$$f_i g_j = f_j g_i$$

for all $0 \le i, j \le n$. Equivalently, there exists an $h \in \bar{K}(V_1)^*$ such that $g_i = h f_i$ for every i. We write

$$\phi : V_1 \dashrightarrow V_2, \qquad \phi = [f_0, \ldots, f_n],$$

and, at every point P satisfying the conditions above, set

$$\phi(P) = [f_0(P), \ldots, f_n(P)].$$

If V_1 and V_2 are defined over K, then G_K acts on $\bar{K}(V_1)$ as follows. For $f \in \bar{K}(V_1)$ and $\sigma \in G_K$, define

$$f^\sigma(P) = \sigma\big(f(P^{\sigma^{-1}})\big)$$

whenever the right hand side is defined. The induced action on a rational map is

$$\phi^\sigma = [f_0^\sigma, \ldots, f_n^\sigma].$$

Note that $\phi(P)^\sigma = (\phi^\sigma)(P^\sigma)$ for all $\sigma \in G_K$ and all $P \in V_1$ at which the two sides are defined. (In particular, note that we have shifted to writing group actions as superscripts.)

If in addition there is some $h \in \bar{K}(V_1)^*$ such that $hf_0, \ldots, hf_n$ are all in $K(V_1)$, then ϕ is said to be defined over K. By Galois descent (equivalently, Hilbert's Theorem 90), ϕ is defined over K if and only if $\phi = \phi^\sigma$ for all $\sigma \in G_K$, where the equality is equality of rational maps, not necessarily equality of the chosen coordinate tuples.

Remark 1.2. A rational map $\phi : V_1 \dashrightarrow V_2$ is not necessarily well-defined at every point of V_1. But it may be possible to evaluate $\phi(P)$ at a point P where some f_i is not regular, i.e. it is not in $\mathcal{O}_{V_1,P}$, by replacing all the f_i by gf_i for some $g \in \bar{K}(V_1)^*$.

Definition 1.3. A rational map $\phi = [f_0, \ldots, f_n] : V_1 \dashrightarrow V_2$ is *regular or defined* at $P \in V_1$ if there is a function $g_P \in \bar{K}(V_1)^*$ such that

- each $g_P f_i$ is regular at P;
- there exists some i for which $(g_P f_i)(P) \neq 0$.

If such a g_P exists, we set

$$\phi(P) = [(g_P f_0)(P), \ldots, (g_P f_n)(P)].$$

This value is independent of the choice of g_P and of the representative of the equivalence class defining ϕ. A rational map that is regular at every point $P \in V_1$ is called a *morphism*.

This 'clearing of denominators' technique also allows us to give an equivalent alternate formulation.

Proposition 1.4 (Polynomial description of a rational map)**.** *Let X denote the vector or tuple (or just the corresponding elements) $(x_0, \ldots, x_m)$ depending on the context. Every rational map $\phi : V_1 \dashrightarrow V_2$ may be represented in the form*

$$\phi = [\phi_0(X), \ldots, \phi_n(X)],$$

where $\phi_i(X)$ for $i = 0, 1, \ldots, n$ are homogeneous polynomials in $\bar{K}[X]$, not all in $I(V_1)$, having the same degree, and having the property that, for every homogeneous polynomial $F \in I(V_2)$, we have

$$F(\phi_0, \ldots, \phi_n) \in I(V_1).$$

Two such tuples $[\phi_0, \ldots, \phi_n]$ and $[\psi_0, \ldots, \psi_n]$ represent the same rational map if

$$\phi_i\psi_j \equiv \phi_j\psi_i \pmod{I(V_1)}$$

for all $0 \leq i, j \leq n$.

A rational map $\phi = [\phi_0, \ldots, \phi_n] : V_1 \dashrightarrow V_2$ as above is regular (or defined) at $P \in V_1$ if there exist homogeneous polynomials $\psi_0, \ldots, \psi_n \in \bar{K}[X]$ such that

- *$\psi_0, \ldots, \psi_n$ have the same degree;*
- *$\phi_i\psi_j \equiv \phi_j\psi_i \pmod{I(V_1)}$ for all $0 \leq i, j \leq n$;*
- *$\psi_i(P) \neq 0$ for some i.*

If this occurs, we set

$$\phi(P) = [\psi_0(P), \ldots, \psi_n(P)].$$

A rational map that is regular everywhere is called a morphism.

Example 1.5. Let $\phi : \mathbb{P}^m \dashrightarrow \mathbb{P}^n$ be a rational map. Since $\bar{K}[X]$ is a Unique Factorization Domain, we may assume that the ϕ_i's have no common nonconstant factor. Then ϕ is regular at $P \in \mathbb{P}^m$ if and only if some $\phi_i(P) \neq 0$. Hence, ϕ is a morphism if and only if ϕ_i have no common zero in $\mathbb{P}^m$.

These are the definitions, but how do we think of rational maps? A rational map may fail to be defined at a point for two reasons: some of the rational functions in a chosen representation may fail to be regular there, or all the resulting regular coordinates may vanish simultaneously. The set on which a rational map is regular is a nonempty open subset $U \subset V_1$ and hence, since V_1 is irreducible, it is dense in the Zariski topology. On this open subset the rational map is given by regular coordinate expressions and behaves as an algebraic map to V_2.

Suppose $\phi : V_1 \dashrightarrow V_2$ and $\psi : V_2 \dashrightarrow V_3$ are rational maps. Their composition $\psi \circ \phi$ is defined as a rational map provided there is a nonempty open subset of V_1 on which ϕ is defined and whose image contains a point at which ψ is defined. On the nonempty open subset where both maps can be evaluated, we set

$$(\psi \circ \phi)(P) = \psi(\phi(P)).$$

Two rational maps are equal if they agree at every point of some nonempty open subset on which both are defined.

Definition 1.6. Let V_1 and V_2 be projective varieties. We say that V_1 and V_2 are isomorphic if there are morphisms $\phi : V_1 \to V_2$ and $\psi : V_2 \to V_1$ such that $\psi \circ \phi$ and $\phi \circ \psi$ are the identity maps on V_1 and V_2 respectively. We say that V_1/K and V_2/K are isomorphic over K if ϕ and ψ can be defined over K.

A rational map $\phi : V_1 \dashrightarrow V_2$ is a *birational map* or *birational equivalence* if there exists another rational map $\psi : V_2 \dashrightarrow V_1$ such that the compositions are defined as rational maps and

$$\psi \circ \phi = \mathrm{Id}_{V_1}, \qquad \phi \circ \psi = \mathrm{Id}_{V_2}.$$

Consequently, a birational map induces an isomorphism between a nonempty open subset of V_1 and a nonempty open subset of V_2 and conversely, an isomorphism between nonempty open subsets of V_1 and V_2 induces a birational map between V_1 and V_2. In this case, we say that V_1 and V_2 are birationally equivalent.

Proposition 1.7. *Two varieties V_1 and V_2 are birationally equivalent if and only if their function fields $\bar{K}(V_1)$ and $\bar{K}(V_2)$ are isomorphic as $\bar{K}$-fields.*

You are highly encouraged to prove this proposition as a test of your understanding of the passage between algebra and geometry.

Corollary 1.8. *Birationally equivalent varieties are equidimensional.*

Proposition 1.9. *Any irreducible n-dimensional projective algebraic variety is birationally equivalent to a hypersurface in a projective space $\mathbb{P}^{n+1}$.*

Again, you are highly encouraged to prove this proposition as a test of your understanding of the passage between algebra and geometry. You should use the previous proposition along with your knowledge of transcendence degrees and primitive element theorem. In positive characteristic, choose a separating transcendence basis so that the remaining finite extension is separable before applying the primitive element theorem.

Example 1.10. Let the characteristic of K be different from 2, and let V be the projective variety corresponding to $X^2+Y^2=Z^2$. Then consider $\phi : V \dashrightarrow \mathbb{P}^1$ given by $\phi = [X+Z, Y]$. Then it is regular at every point except possibly at a point where $X+Z=Y=0$, which is the point $P=[1,0,-1]$. But we can use

$$(X+Z)(X-Z) \equiv -Y^2 \mod I(V)$$

to see that

$$\phi = [X+Z, Y] = [X^2-Z^2, Y(X-Z)] = [-Y^2, Y(X-Z)] = [-Y, X-Z].$$

These are equalities of rational maps in the sense of the equivalence relation on coordinate tuples. Thus $\phi([1,0,-1]) = [0,2] = [0,1]$, so that ϕ is defined at every point of V. Consequently, ϕ is a morphism. The map $\psi : \mathbb{P}^1 \to V$ given by the formula

$$\psi = [S^2-T^2, 2ST, S^2+T^2]$$

is a morphism and in fact gives an inverse for ϕ, so V and $\mathbb{P}^1$ are isomorphic. (Can you geometrically explain the visible relation between the Pythagoras parametrization and the above morphisms?)